

Restoration of Endodontically Treated Teeth

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Restoring teeth functionally and esthetically after end-odontic treatment often presents a challenge to the dentist. Typically, root canal treatment is initiated because of deep caries or trauma, both of which often result in extensive loss of tooth structure. Additional tooth tissue is removed for endodontic access, cleaning and shaping of the root canal, and post space preparation, further reducing the structural integrity of the tooth and decreasing its resistance to fracture.[1,2] At one time it was believed that endodontically treated teeth are inherently more brittle and susceptible to fracture.[3,4] Subsequent research has shown that the dentin of endodontically treated teeth exhibits mechanical properties equivalent to that of untreated teeth.[5,6]

It has also been proposed that a portion of the sensory feedback mechanism is lost when the neurovas-cular tissue has been removed from the tooth in the course of endodontic therapy, an effect confirmed in an in vivo study.[7] Clinically, this means that the patient can inadvertently bite with more force on an end-odontically treated tooth than on a vital tooth because of the impaired sensory feedback mechanism.

This may account for the failure of endodontically teeth that would otherwise appear to be fracture resistant (Fig 21-1).

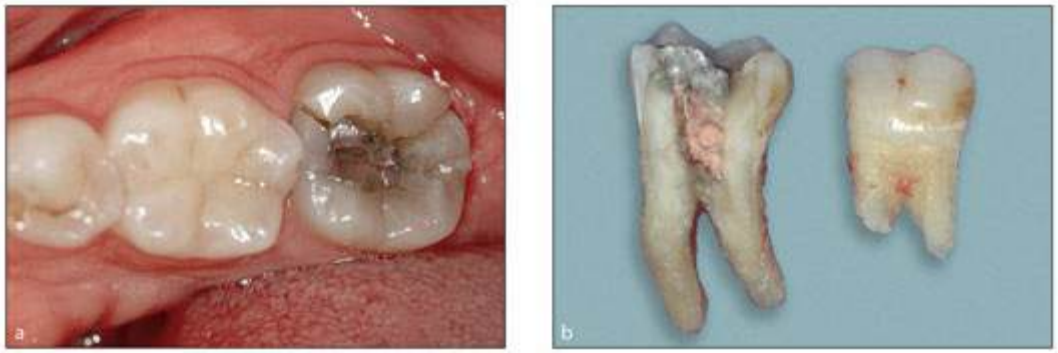


Fig 21-1*(a and b)* Unrestorable, fractured, endodontically treated mandibular second molar with an occlusal amalgam restoration.

While endodontically treated teeth may fail for a variety of reasons—eg, caries, endodontic failure, and periodontal disease, among others—tooth fracture remains a significant factor in the posttreatment extraction of these teeth.[8] The choice of definitive restoration plays an important role in survivability. In selected instances, endodontically treated teeth can be restored with a directly placed restoration, particularly in the anterior region. However, restoring a severely damaged tooth to function and esthetics often requires the use of a full-coverage restoration. Providing adequate resistance and retention form when there is a minimal amount of remaining tooth structure can be problematic. When a moderate amount of tooth structure has been lost, the tooth may be built up with restorative material such as amalgam or composite to provide a *core* to support and retain the crown. When the damage is severe and there is inadequate tooth structure to retain the core, other methods of retention must be employed. This may consist of adhesive bonding techniques, providing auxiliary mechanical retention, and/or placement of a *post* into the root canal that extends into the core and retains it.

The longevity of endodontically treated teeth is difficult to evaluate because of the many possible variables. These include patient factors, such as systemic health, periodontal condition, and caries susceptibility; tooth-related factors, such as function and the amount of tooth structure remaining; as well as material factors, such as the type of post, cement, core, and definitive restoration. While the amount of clinical information grows each year, there are currently insufficient data to predict the effect of each of these factors on the success or failure of an individual tooth. However, analysis of large data sets from insurance companies indicates that endodontic treatment followed by restoration is very successful in terms of tooth retention. Salehrabi and Rotstein[9] analyzed the outcome of endodontic treatment in a

¼6 million cohort of US patients and found an 8-year survival rate of 97%. Chen et al[10] conducted a similar study on a Taiwanese data set of 1.56 million patients and found a tooth retention rate of 93% after 5 years.

Risk Analysis

The survivability of the individual tooth and its ability to fulfill its intended function must be carefully considered, and the quality of the existing endodontic treatment should be assessed. The absence of symptoms and of periapical pathology, as well as the presence of a radiographically dense and adequately extended obturation, should be confirmed. Periodontal disease should be controlled, and the crown-root ratio should be assessed in view of the position of the tooth in the arch and its anticipated function. Success is enhanced when a restored endodontically treated tooth is in proximal contact on both the mesial and distal.[11] If the root canal filling has been exposed to the oral environment because of leakage or loss of a restoration, contamination of the periapex with bacteria or toxins may occur within days.[12,13] Retreatment should be considered in situations where the root canal filling has been exposed to the oral environment. There are no clinical data available to give specific guidance on how long the root canal filling may be exposed before retreatment should be accomplished, but in vitro studies suggest a range of days to several months.[12,14] Because of the difficulty and expense involved in retreatment after restoration, a conservative approach is recommended. Many clinicians use 2 weeks of exposure as an arbitrary guideline for retreatment. Because of the risk of contamination, coronal sealing of the root canal system should be accomplished as soon as possible after completion of the root canal therapy.[15] A long time span between the endodontic treatment and placement of the definitive restoration leads to a significantly higher failure rate.[16,17]

Perhaps the most important factor related to the survivability of an endodontically treated tooth is the amount of coronal and radicular tooth structure that remains prior to restoration. This factor appears to be of paramount importance and will be considered in detail later in this chapter. The tooth structure that will be available following tooth preparation should be visualized and should be assessed for the provision of adequate retention and resistance form for the subsequent crown. Key to this factor is the presence of an adequate ferrule. The term *ferrule* describes the band of circumferential material that encircles the cervical area of a tooth restored with a crown.[18] As commonly used in dentistry, *ferrule* also refers to the dimension

of remaining vertical tooth structure from the anticipated margin of the preparation to its coronal extent, which is available for encirclement by the crown. The "ferrule effect" serves to reinforce the remaining tooth structure and helps the restoration withstand lateral forces.[19]

The ferrule provides resistance form and enhances longevity.[20] Ferrule length was found to be more important than post length in increasing fracture resistance to cyclic loading of crowned teeth.[21] Indeed, some authors have noted that the effect of the remaining dentin height is so significant as to make the effect of the type of post and core employed irrelevant with respect to tooth survival.[22] Lack of an adequate ferrule increases the risk of failure because of post-crown dislodgment, post fracture, and root fracture.[23,24] Sorensen and Engelman$_{19}$ noted that a ferrule of only 1.0 mm in vertical height doubles the resistance to fracture of a tooth without remaining coronal tooth structure. Optimal beneficial effect can be expected from a ferrule of at least 1.5 to 2.0 mm of vertical height.[18,25]

Adequate sound tooth structure should be present to allow placement of a restoration margin that will not encroach on the biologic width, which will ensure continued periodontal health. *Biologic width* has been defined as "the dimension of the junctional epithelium and connective tissue attachment to the root above the alveolar crest" and has been estimated to be approximately 2 mm.[26] Violation of this dimension with a restorative margin may result in persistent periodontal inflammation.[27] Given the requirement of a minimal ferrule height of 2 mm, at least 4 mm of sound supra-alveolar tooth structure is necessary for mechanical stability and periodontal health.

If there is insufficient remaining tooth structure to adequately retain a core, the suitability of the existing root canals to receive a post should be assessed. A radiographic assessment of the availability of a sufficient length of straight canal of adequate diameter should be made preoperatively and factored into the prognosis. Periapical radiographs taken from multiple angles are often helpful.

There are few teeth that cannot be endodontically treated and restored to some level of function provided the dentist has the skills and the patient is willing and able to commit the time and resources. Current techniques provide a high degree of success for endodontic treatment. Even teeth with no tooth structure remaining above the osseous crest can be restored if techniques such as forced orthodontic eruption$_{28}$ or crown lengthening surgery are utilized to create more available vertical tooth structure for crown retention. However, these techniques add expense, patient inconvenience, and discomfort and may compromise esthetics and support of

adjacent teeth.[29] In addition, because roots are tapered, these procedures result in a decrease in the cross-sectional dimension of the final preparation and also decrease the crown-root ratio. With the ever-increasing success and availability of implant treatment, the concept of "restorability" has changed. The tooth may be indeed be restored, but what is the anticipated longevity? A realistic assessment must be made of the long-term prognosis of the restored tooth compared with that of an implant-retained restoration.

The use of osseointegrated implants for the replacement of teeth offers an alternative for teeth with a questionable prognosis. A systematic review of clinical studies assessing the survival rate of single-tooth implants with teeth that had received root canal treatment followed by coronal restoration concluded that there is no difference in the survival outcomes of these two treatment modalities.[30] Fifty-five studies related to single-tooth implants met the criteria for inclusion in the analysis. Success rates ranged from 89% to 100%. The largest study analyzed, a prospective study assessing 2,900 implants, indicated a success rate of 98% at 3 years.[31] Thirteen studies related to the survival of endodontically treated teeth with coronal restorations met the review criteria and were analyzed. Survival ranged from 81% to 100%. The largest study included, with 19,817 teeth, showed a survival rate of 97% at 72 months.[32]

The conscientious practitioner will examine the risk factors associated with the successful endodontic treatment and restoration of a marginal tooth versus extraction and the placement of an implant. This decision involves expertise in implant treatment planning, which is beyond the scope of this text. In making the decision, many factors must be considered, such as the quantity and quality of remaining bone, esthetic and soft tissue considerations, and of course, the desires and medical status of the patient.

Treatment Planning

The type of definitive restoration required for an endodontically treated tooth is primarily determined by the amount of remaining tooth structure, the position of the tooth in the arch, and the functional forces placed on it. An appropriate restoration may range from a direct resin composite in an anterior tooth to a full crown. The greater the amount of tooth structure lost, the greater will be the need for additional retention for the core, often in the form of a post. The ability of a patient to exert heavy occlusal forces, which might displace a core and the crown it supports,

decreases from the posterior regions of the dental arch to the anterior.[33] However, the direction of the functional forces transitions from primarily vertical (compressive) in the posterior region to lateral (tensile or shear) in the anterior. These tensile forces tend to displace the crown/ core, and therefore the need for additional resistance form is greater in the anterior region, and a post is often required. In the posterior region, although the forces are greater, they are primarily vertical, and sufficient retention and resistance form for the core is often obtained by extending it into the pulp chamber. For these reasons, a post is rarely required in a molar tooth. However, when a restored tooth is to be used as an abutment for a removable or fixed partial denture, greater lateral forces should be anticipated, and the additional resistance form provided by a post should be considered.[34–36] Because of the differences in magnitude of functional forces, direction of forces, and anatomical differences, anterior and posterior teeth present different restorative challenges.

Anterior teeth

In the anterior region, there will frequently be sufficient remaining tooth structure to restore the tooth to function and esthetics with a direct restoration. In cases of traumatically induced pulpal necrosis, the only restoration required is to seal the access preparation (Fig 21-2). In other situations, there may be moderately sized Class 3 cavities in addition to the access. In these cases, neither a post nor a full-coverage restoration is required, and the tooth can be restored with bonded resin composite. When a moderate amount of coronal tooth structure is missing but at least 50% of the coronal enamel remains, the bonded porcelain veneer may be the restoration of choice (Fig 21-3). A tooth conservatively prepared for a porcelain veneer allows for the creation of excellent esthetics without compromising the strength of the remaining tooth structure. Laboratory studies suggest that endodontically treated incisors restored with porcelain veneers have similar resistance to fracture as an intact tooth.[37,38] There is no need for post placement in these situations. Although some laboratory studies[39,40] have indicated that a metallic post strengthens an intact anterior endodontically treated tooth, the majority of studies have suggested that the fracture resistance of these teeth is not affected by placement of a post.[41,42] In addition, placement of a post in an anterior tooth with an intact incisal edge is often complicated by a lack of draw for the post into the canal through the access opening. Removing additional tooth structure in order to place a post is not indicated.



Fig 21-2

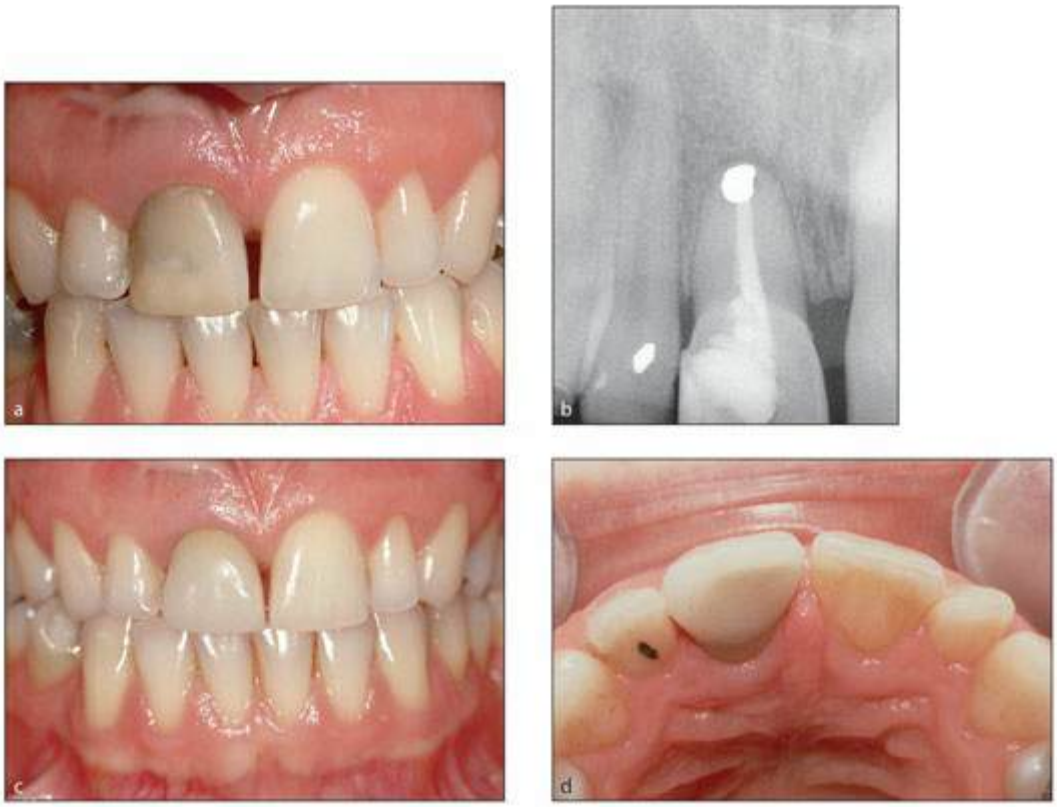


Fig 21-3*(a)* Preoperative view of a maxillary right central incisor. *(b)* Preoperative radiograph. *(c and d)* Seven-year postoperative views of the endodontically treated maxillary right central incisor restored with a porcelain veneer.

When the decision has been made to restore the tooth with a crown because of loss of structural integrity or esthetic demand, an assessment must be made as to whether there is sufficient tooth structure remaining to retain the core or whether retention and resistance form must be augmented with a post. Because of the

predominantly lateral forces that the anterior teeth are subjected to in function, careful consideration should be given to the placement of a post. In some cases, the dentist may wish to defer this decision until after preparing the tooth. An assessment may be made at that time whether sufficient vertical walls remain to resist the lateral forces that the crown will be subjected to without the placement of a post. Because of the size of the canines and the maxillary central incisors, a post can often be avoided in these teeth, and adequate retention and resistance form can be provided by the placement of a core buildup. With maxillary lateral incisors and mandibular central and lateral incisors, providing adequate tooth reduction for the crown often results in very little remaining tooth structure, and the need for a post is clear.

The canals of maxillary and mandibular canines and maxillary central incisors are usually straight and of adequate size and shape for post placement. On the other hand, the canals of mandibular central and lateral incisors, and often maxillary lateral incisors, are small and ribbon shaped and offer variable opportunity for post placement.

Posterior teeth

Clinical studies have demonstrated that a key element in the successful restoration of endodontically treated posterior teeth is the placement of a full coronal cuspal-coverage restoration. In a retrospective study of 1,273 endodontically treated teeth, Sorensen and Martinoff[43] found that a full-coverage restoration increased the success rate of posterior teeth. In another retrospective study, Aquilino and Caplan[44] showed that end-odontically treated molars not crowned after obturation were lost at a rate six times greater than teeth that were crowned. In a systematic review of clinical studies of endodontically treated teeth, Stavropoulou and Koidis[45] determined that the 10-year survival rate of crowned teeth is 81% versus 63% for direct restorations. In a study of molar teeth, Nagasiri and Chitmongkolsuk[46] found that the survival rate of uncrowned teeth was 96% at 1 year, 88% at 2 years, and 36% at 3 years.

Examining the status of the endodontically treated teeth extracted during the period evaluated in two large epidemiologic studies reinforces the importance of cuspal coverage. In the previously cited study by Salehrabi and Rotstein,[9] 85% of extracted teeth had no full coronal coverage. In another study, teeth restored with amalgam or composite showed an incidence of extraction of 6.3%; teeth with full coronal restoration had an incidence of only 2.5%.[32]

Interestingly, despite compelling evidence to crown such teeth, many teeth that

receive endodontic treatment are not crowned. Scurria et al[47] followed up the restorative outcome of 1,193 endodontically treated teeth through an insurance claims system and reported that the percentage of noncrown restorations placed during the subsequent 2 years after end-odontic therapy was 50% in the molar group, 54% in the premolar group, and 67% in the anterior tooth group.

Regardless of the seemingly poor performance of direct restoration of endodontically treated posterior teeth, it should be cautiously considered in cases where substantial tooth structure remains. Access preparation has little effect on the stiffness of endodontically treated maxillary premolars, but the removal of the marginal ridges with a larger preparation has a significant effect.[48] The success rate of direct restorations is improved proportionately to the amount of tooth structure remaining$_{46}$ and by using bonded resin composite.[49] A short-term (30-month) clinical evaluation of the use of fiber-reinforced composite (FRC) posts and direct resin composite restorations revealed no failures within that time period.[50] A 3-year clinical study concluded that the clinical success rates of endodontically treated premolars restored with FRC posts and direct composite restorations were equivalent to full coverage with metal-ceramic crowns.[51]

In the face of confusing and contradictory data regarding full-coverage versus direct restoration of endodontically treated teeth, the clinician should take a conservative approach. The preponderance of clinical data suggests that the full coronal coverage restoration is the more predictable and long-lasting restoration. A direct resin composite restoration may be considered when there is substantial remaining tooth structure and lower functional loads are present. However, it should be realized that the strengthening effect of the bonded resin composite may be lost over time. It has been shown that the effect diminishes significantly with both thermal cycling[52] and functional loading[53] of the restoration. Degradation of the resin-dentin bond over time has also been demonstrated in vivo.[54] Therefore, the long-term strengthening effect of the intracoronal bonded restoration must be questioned, and the clinician should choose this treatment option with caution.

Restoration of endodontically treated teeth with amalgam without cuspal overlay has a low clinical success rate and cannot be recommended.[55] However, in the hands of a skilled operator, a cuspal-coverage amalgam restoration may provide a low-cost alternative to a crown (Fig 21-4). In 1980, Nayyar et al[56] described the *amalcore*, or coronal-radicular, restoration. Rather than placement of a post, the pulp chamber and the coronal 2 to 3 mm of each canal are used for retention of the buildup material (Fig 21-5). A post is generally not needed with the amalcore.

Although some authors have found that placement of a post increases the resistance of the amalcore to fracture,[57,58] others have found no difference.[59] A post is required only when insufficient pulp chamber depth is present to retain the core (Fig 21-6). In a retrospective clinical study of more than 400 coronal-radicular restorations, no failures were noted that could be attributed to the core buildup. If appropriate thickness of occlusal amalgam can be provided and adequate axial contours, proximal contacts, and occlusion can be developed, the amalcore technique can be used to create a long-lasting definitive restoration. A 6-year clinical study demonstrated the effectiveness of complex, cusp-capping amalgam restorations mechanically or adhesively retained.[59]

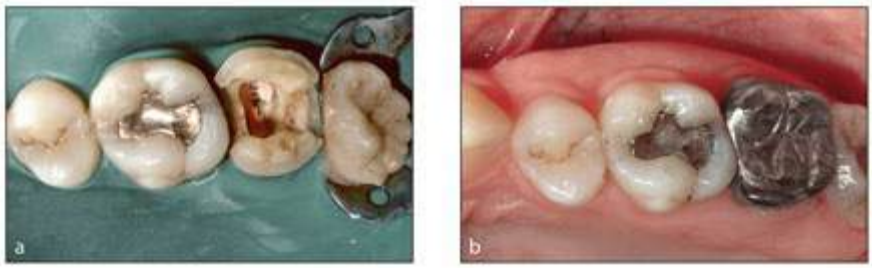


Fig 21-4*(a)(b)*



Fig 21-5 Amalcore with adequate chamber retention.



Fig 21-6

Maxillary premolars are a unique subset of posterior end-odontically treated teeth. Anatomically, the root is often small in diameter relative to the dimensions of the crown to be supported, and there may be little tooth structure to retain the core. The pulp chamber of these teeth is typically small and provides little mechanical retention for the core. A post is indicated in these cases. In addition, these teeth are subjected to a mixture of shear and compressive forces, and the need for a post to provide additional resistance form is often similar to that for an anterior tooth. Similarly, lateral forces exerted on a maxillary premolar functioning as the abutment for a removable partial denture often necessitate a post for additional resistance (Fig 21-7).



Fig 21-7 *(a)* Preoperative view of an endodontically treated maxillary second premolar. *(b)* Maxillary second premolar after restoration with a tapered prefabricated post and amalgam core. The canal was not enlarged for post placement.

Maxillary premolar roots are often fluted and may be curved, presenting a high risk of strip perforation during preparation of the post space.[60] The palatal root has been shown to have a larger and straighter canal than the buccal and should be utilized for post placement.[61]

For the molar tooth, the decisions are more clear-cut. The forces on posterior teeth are predominately vertical. Therefore, development of additional resistance form to withstand lateral forces is not as commonly needed as in anterior teeth. Because of the morbidity associated with post placement (Fig 21-8), a post is indicated in a posterior tooth only when other more conservative retention and resistance features cannot be used for the core. These features include chamber/canal orifice retention, amalgam pins, and threaded pins, all of which have been shown to be exceedingly effective.[62] When a post is deemed necessary, the palatal canal of maxillary molars and the distal canal of mandibular molars are usually the best choice.



Fig 21-8 Mandibular molar with post perforation in the mesial concavity of the distal root.

Material Selection

Core material

A core is routinely placed in teeth for which a full-coverage indirect restoration is indicated. The core replaces lost tooth tissue and augments retention, resistance, and support for the crown. Restoring a crown preparation to ideal height through the use of a core increases the ability of the crown to withstand the fatigue forces that might displace it.[63] The core is retained through the use of adhesive bonding, existing mechanical retentive features, a cemented post, or with a combination of methods.

A variety of directly placed restorative materials are available for use as a core with or without a post; amalgam, glass ionomer (both conventional and resin-modified), and resin composite are popular choices. Desirable characteristics of a core material are listed in Box 21-1.

Box 21-1	Characteristics of the ideal core material
• Stability in a wet environment • Ease of manipulation • Rapid, hard set for immediate crown preparation •High compressive strength •High tensile strength •High modulus of elasticity (rigidity) •High fracture toughness •Inert (no corrosion) •Cariostatic properties •Biocompatibility •Low cost	

Amalgam has numerous advantages as a core material. Its high strength has been confirmed in laboratory studies under both static and dynamic loads.[64,65] The modulus of elasticity of amalgam is three times that of a typical resin composite and thus provides excellent support for all-ceramic crowns.[66] It is easy to place, it is relatively insensitive to moisture contamination, and the color contrast facilitates margin placement. These factors make amalgam an excellent choice when the clinician is faced with placing a core deep within the gingival sulcus. On the other hand, amalgam has a number of disadvantages. Its early strength is low, necessitating a 15- to 20-minute wait until the buildup can be prepared for the crown, even when fast-setting spherical alloy is used. It is messy to prepare and can result in irreversible staining of the marginal gingiva during preparation. Therefore, amalgam should not be used as a core material in a tooth in the esthetic zone.

Conventional glass-ionomer materials have the advantages of fluoride release, ease of manipulation, natural color, biocompatibility, corrosion resistance, and dimensional stability in a wet environment.[67] They are also offered in a metal-reinforced formulation that offers color contrast to tooth structure. However, they have the major disadvantage of brittleness and low resistance to fatigue forces, which makes them unsuitable for use in extensive cores.[64,65] Resin-modified glass-ionomer materials offer improvements over the conventional glass ionomers but still fall short compared with bonded resin composite in terms of fracture toughness, bond strength, and dimensional stability.[68–70]

Resin composite is the most popular core material because of its excellent mechanical properties, bond to tooth structure when used with an adhesive, and ease of use. It is available in light-curing, dual-curing, and self-curing formulations. It is provided as both a tooth-colored material to be used as a core material under anterior all-ceramic restorations and as a color-contrasting material to be used as a core material under metallic or opaque ceramic (alumina or zirconia) restorations. Resin composite has a fracture toughness[68] and transverse strength similar to that of amalgam.[66] Resistance to fatigue loading is intermediate between glass ionomer and amalgam.[64,65] The dimensional stability of resin composite in a wet environment is greater than that of glass-ionomer materials but less than that of amalgam.[70,71] The clinical significance of this property is unclear. One study found that water storage of extracted teeth with composite cores resulted in expansion, which decreased the fit of the subsequent crown. Amalgam was found to be unaffected.[71] Another study found that water storage affected the fit of crowns on preparations of natural tooth as well as composite and that the placement of a well-fitting interim restoration

improved the fit of both.[72]

In general, the bond of resin composite to the dentin of the pulp chamber is not as strong as that to coronal dentin.[73,74] And, as previously noted, the resin bond may weaken with time and function in the oral environment. For these reasons, the clinician should always strive to create additional mechanical retentive features to retain the resin composite core when possible.

Ribbon reinforcement

The fabrication of an individualized, directly fabricated post and core utilizing adhesively bonded woven polyethylene or glass ribbon and resin composite has been suggested. Examples of these materials include everStick (Stick Tech), which consists of silanated glass fibers impregnated with polymethyl methacrylate (PMMA) and bisphenol glycidyl methacrylate (bis-GMA), and Ribbond (Ribbond), with ultrahigh–molecular weight polyethylene fibers (UHMWP) in a cross-linking lock-stitch leno weave. It has been proposed that these fiber ribbons increase retention and prevent crack propagation through the core.[75] The clinical technique involves embedding the ribbon into resin composite placed in the root canal and within the core and has been described in detail in the literature.[76,77] Newman et al[78] tested the fracture resistance of a variety of post constructs loaded at a 45-degree angle in extracted teeth. When Ribbond and resin composite were used as a post and core in a conservatively prepared root canal, the fracture strength was significantly less than that observed with fiber posts. When tested in flared, thin-walled canals, the fracture strength was equivalent to that with fiber posts.[78] One clinical study showed a high survival rate of teeth restored with this technique.[79] However, a 6-year prospective study showed an overall failure rate of the everStick post and core of 38.7%. This was found to be superior to no post at all but inferior to the 23.4% failure rate seen with a bonded FRC post.[24]

Selection of a post

A wide variety of post designs and materials are available to the clinician (Fig 21-9). Posts may be characterized by manufacturing process (custom-cast versus prefabricated), engagement with root dentin (active versus passive), shape (parallel-sided versus tapered), surface features (smooth versus macroretentive), and material (metallic versus ceramic versus FRC). While each design has its advocates, three fundamental principles should guide selection:

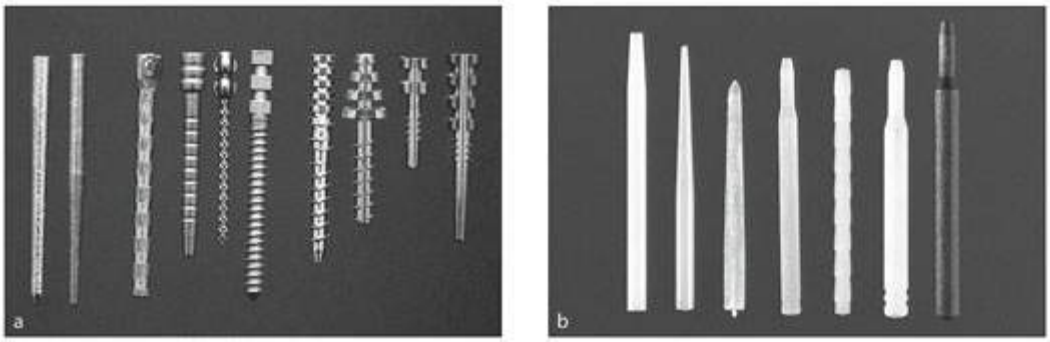


Fig 21-9*(a)* Prefabricated posts. *(left to right)* Passive tapered: Endowel (Star), Filpost (Ivoclar Vivadent); passive parallel: Parapost Plus (Coltène/Whaledent), BCH (3M ESPE), Unity (Coltène/Whaledent), Boston Post (Roydent); active: Flexi-Post (Essential Dental Systems), V-Lock (Brasseler), Radix (Dentsply Maillefer), Cytco (Dentsply Maillefer). *(b)* Nonmetallic posts. *(left to right)* CosmoPost (Ivoclar Vivadent), CeraPost (Brasseler), Luscent Anchor (Dentatus), Light-Post (Bisco), FibreKor Post (Pentron), Aestheti-Plus (Bisco), and C-Post (Bisco).

1. Tooth structure should be preserved to maintain fracture resistance. Ideally, the post should fit the canal shape produced by the endodontic procedure without further removal of tooth structure. Fulfillment of this principle typically necessitates the use of a tapered post.
2. The post should not increase the susceptibility of the tooth to root fracture but rather reduce its incidence, if possible. Selection of a post that fits the canal passively and provides physiologic stress distribution fulfills this requirement.
3. The post should provide adequate retention of the core-crown assembly without compromising the principles of conservation of tooth structure and preservation of root integrity. This principle is satisfied primarily through the use of a post of adequate length and secondarily by the use of adhesive bonding techniques.

No single post system fulfills these principles for all teeth. Clinical judgment must be exercised to select a post system or core-retention method appropriate for each individual situation.

Post Designs and Materials

Cast metal post and core

The custom-cast post and core has a long history of success in restorative dentistry. However, laboratory studies[80–82] have consistently shown that the fracture resistance of teeth restored with custom-cast posts is lower than that of teeth restored with

many different prefabricated posts. In addition, retrospective clinical studies[23,43] have shown metallic prefabricated parallel-sided posts to have greater clinical success than custom-cast posts. This, coupled with the added expense and extra appointment required to fabricate the custom-cast post, makes its routine use unpopular. In addition, there are concerns about leakage of the interim post crown allowing reinfection of the root canal system during the temporization period.[83]

There are several circumstances in which the custom-cast post is the post of choice:

- When multiple post and core restorations are planned in the same arch, the laboratory-fabricated custom-cast post may be the most time- and cost-efficient method. The teeth are prepared for the posts, and the final crown preparations are completed so that all crown margins are on tooth structure. It is important that the crown preparation be completed before the impression for the post and core is made so that the axial contours of the core can be fabricated correctly.

  An impression is made with an elastomeric impression material used with an injection technique, which allows the impression material to flow into the total length of the prepared canal space (Fig 21-10). This can be best accomplished using one of the following techniques. A 25-gauge needle is placed into the canal before the impression is made. The syringe material is then injected into the canal until it begins to flow out the top of the orifice. While the syringe material is being injected, the needle is slowly removed from the canal. The needle serves as an escape channel for the trapped air and allows the elastomeric impression material to reproduce the entire length of the canal space. Alternatively, the syringe material may be placed directly into the canal with a needle tube, which is attached to the syringe. The tip of the needle tube must be long enough to extend to the entire length of the prepared canal space. The elastomeric material is injected as the needle tube is slowly removed from the canal. A final technique that can be used to ensure that impression material flows into the entire length of the prepared canal is the use of a lentulo spiral (see Fig 21-13). The syringe impression material is initially injected into the canal until it flows out of the orifice. The lentulo spiral is then introduced into the canal and rotated at low rpm to force the impression material apically. The lentulo spiral should be slowly and gently advanced until the apical aspect of the prepared canal is felt and then slowly removed while it continues to rotate. No reinforcement of the impression material in the canal space is required with the newer impression materials. The impression is poured, and the custom posts are fabricated in the laboratory. At a

subsequent appointment, the posts are cemented and the final impression for the restorations is made without further tooth preparation.

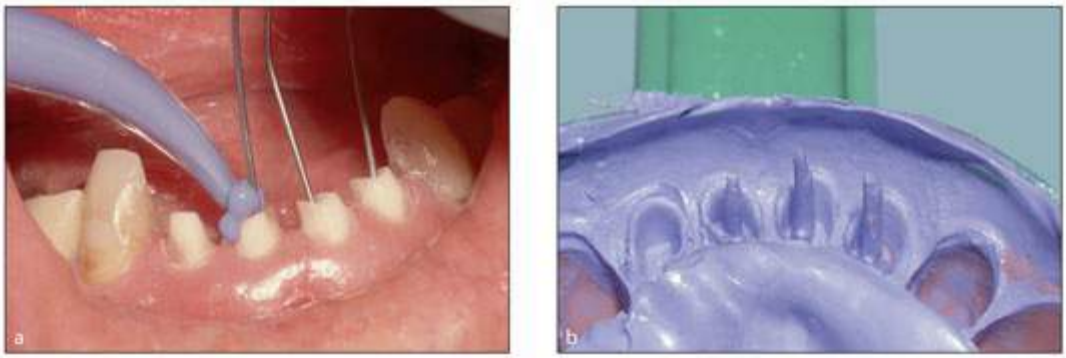


Fig 21-10*(a)* Use of 25-gauge needles to allow air to escape from canals during the impression making to ensure a complete impression of the canal spaces. *(b)* Final impression of canal spaces for laboratory fabrication of custom-cast posts.

- When a small tooth, such as a mandibular incisor, requires a post and core, a prefabricated post may be difficult to use. Commonly, there is minimal space around the post for the core buildup material. In this situation, the custom-cast post serves well.
- Occasionally, the angle of the core in relation to the root must be altered. It is not advisable to bend prefabricated posts; therefore, the custom-cast post and core most successfully fulfills this need (Fig 21-11).

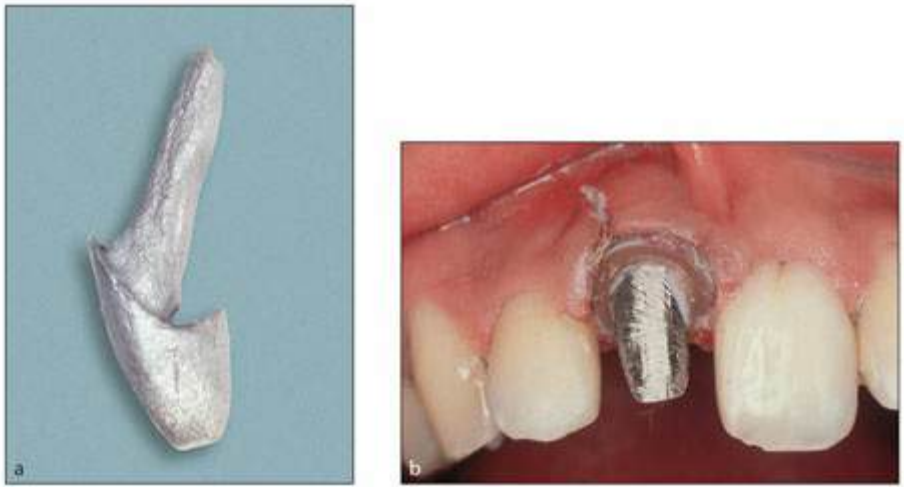


Fig 21-11*(a and b)* Custom-cast post to allow a change in the angle of the core in relation to the post.

Prefabricated metal posts

Prefabricated posts are available in a variety of metallic compositions and styles. They are most commonly manufactured from stainless steel, but many designs are

also available in titanium alloy. Because stainless steel contains nickel, some clinicians have been concerned about the possible allergenicity of the corrosion products of this metal. Titanium alloy posts were introduced to address this concern. A disadvantage of titanium alloy posts is that they have low radiodensity and may be difficult to detect on radiographs.[84] The passive parallel post, the tapered passive post, and the active post designs will be considered here. The applications for these post types are similar, with certain qualifications as outlined below.

Passive parallel-sided metal post

The prefabricated metallic post by which all other posts are measured has traditionally been the Parapost (Coltène/Whaledent) (see Fig 21-9a). The success of this post style has been demonstrated in the laboratory[21,85,86] as well as clinically.[23,43] Compared with other metallic post designs, the parallel post has been shown to provide the most favorable stress distribution$_{87}$ and exhibits greater retention than the passive tapered post.[88] However, a biologic price must be paid for this increased retention. The naturally tapered canal space must be enlarged to accommodate a parallel post. Enlargement of the canal space is not consistent with the ideal of maintaining as much tooth structure as possible. For this reason, use of the passive parallel post is recommended only when increased retention is needed and the parallel canal preparation will not jeopardize the integrity of the root.

Passive tapered metal post

A goal of all post systems should be minimal removal of tooth structure before post placement. Therefore, the ideal post system requires no further preparation after removal of the gutta-percha. Because the natural shape of the canal is tapered, the passive tapered post best fulfills this criterion. A major advantage of the passive metallic tapered post is that the post can be modified to fit the tapered canal rather than the canal having to be enlarged to fit the post.

The major disadvantage is that the tapered post provides the least retention compared with other post designs. This means that retention must be gained through maximum post length. When the available post space is not long enough to allow for adequate post length, a more retentive post may be indicated. A second commonly stated disadvantage of the tapered post is the alleged wedging effect, which results in increased stress at the apex and possible resultant root fracture.[87,89] This effect has been demonstrated in laboratory studies$_{80}$ with custom-cast tapered post and core restorations. However, this theoretical wedging effect does not appear to be

valid when a passive tapered post is used in conjunction with an acceptable core material and a crown.[90] A 10-year retrospective study confirmed the clinical durability and retentive ability of smooth tapered posts of appropriate length.[91]

An excellent indication for the passive tapered post is in teeth with small canals and thin, fragile roots, such as maxillary premolars (see Fig 21-7). However, posts of this design may be used routinely in teeth with normal canal configuration and sufficient canal length to provide the necessary retention.

Active metal post

An active post is one that engages (screws into) the dentin of the canal space. There are many active post designs available, including those requiring a tap, self-tapping posts, split-shank posts, and hybrid posts, which contain both active and passive features (see Fig 21-9a). It is difficult to generalize about active posts because of their design differences. In general, it has been reported that the active threaded post has the greatest retention, followed by the parallel post, with the tapered post providing the least retention.[85,86,92] Therefore, the post design should be chosen, in part, by the amount of post retention that the clinical situation requires. If the canal length is adequate and the canal configuration is normal, either the tapered or the parallel post may be selected. However, if the length of post space available is minimal or the canal space is funnel shaped, an active post may be required because of the difficulty in gaining adequate axial retention.

Traditionally, the major concern about active posts has been the potential for vertical fracture of the tooth during placement of the post.[85] In a photoelastic study, stress concentration was noted at each thread of the screw post.[93] However, many laboratory studies support the use of the newer generation of active posts.[94–99] It has been shown that the active post should not be "bottomed out" when it is inserted.[98] After complete seating of an active post, it should be unscrewed one-fourth of a turn. This results in decreased residual stress in the root. It has also been shown that, at shorter lengths, the active post produces less stress than other styles of prefabricated posts.[100] Therefore, active posts are indicated when the canal length is insufficient to gain adequate retention with a passive post. Situations in which an active post may be useful include a short canal space (Fig 21-12) or a canal that is partially occluded by a broken instrument or post.

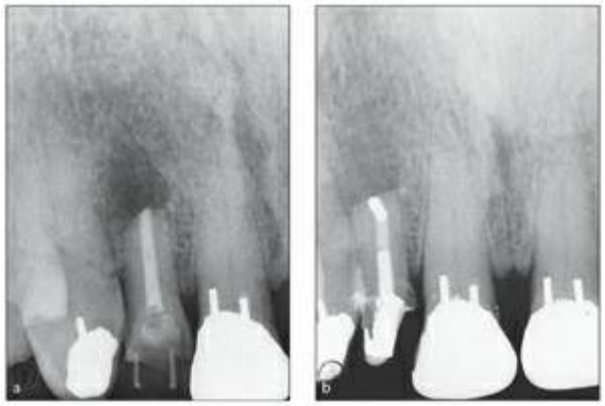


Fig 21-12*(a)* Preoperative view of a maxillary lateral incisor with a short canal space. *(b)* The maxillary lateral incisor after restoration with a Brasseler V-Lock active post and amalgam core.

Caution should be used in selecting an active post. In a prospective clinical study, Schmitter et al[101] compared the clinical performance of a metallic screw post with an FRC post over a 5-year period. The survival rate of the teeth restored with a screw post was 50%, while the teeth restored with an FRC post showed a survival rate of 72%. The predominant failure mode of the screw post group was root fracture, while the failures with the FRC post were due predominantly to debonding.

Cementation of metal posts. Traditional cements for posts and post and core restorations have been investigated extensively.[85,102–107] These include zinc phosphate, polycarboxylate, and glass-ionomer cements. Both zinc phosphate and glass ionomer have been commonly employed because of their ease of use and their history of clinical success. Traditional cements provide retention for the post primarily through a frictional effect. The cement flows into irregularities in the dentin and post surfaces. On hardening, mechanical retention is created.

The lentulo spiral is an excellent instrument for the placement of the traditional cements (Fig 21-13). The cement may also be placed in the canal with a needle tube, as long as the tip of the tube is inserted to the bottom of the canal space *and* the cement is extruded from the tip as it is slowly removed from the canal. After the cement is placed in the canal, the post is coated with the cement and inserted. The post should be inserted slowly to allow dissipation of hydrostatic forces that might impede complete seating. This is a particular issue with precisely fitting, parallel-sided posts.[108] Many prefabricated posts have a venting mechanism incorporated into their design. Tapered posts are considered "self-venting."[109]



Fig 21-13 A lentulo spiral is used to place cement into the canal space. This instrument can also be used to enhance impression material adaptation to prepared canal spaces.

Recent trends are toward adhesive cementation of metallic posts, as well as FRC posts, with resin cement. Two adhesive interfaces are involved in this process: the interface of the cement with the dentin and that of the cement with the post. Regarding the cement interface with the metal post, the bond of resin cement has been shown to be improved by air abrasion and by the use of tribochemical coating.[110] Tribochemical coating involves air abrading the post surface with silicate-coated alumina particles followed by chemical silanization. This chair-side process is available commercially as CoJet (3M ESPE). The interface with dentin will be considered in detail later in this chapter.

Ceramic posts

The all-zirconia posts are white, biocompatible,[111] and radiopaque. When used in conjunction with a resin composite core material, a very esthetic post and core can be fabricated. They are strong compared with prefabricated metal and FRC posts, but they are reported to have low resistance to crack propagation.[112] They have an even higher modulus of elasticity than stainless steel,[113] which creates concerns that stress concentration might predispose the tooth to root fracture.[114] However, in vitro studies have shown that failures predominantly involve fractures of the post without root fractures.[115–117] Possibly, the fracture of the ceramic post absorbed most of the energy, thereby saving the remaining root from fracture.[20] While this might be perceived as a beneficial root-preserving effect, fracture of the ceramic post presents a clinical dilemma in that the remaining portion is extremely difficult to remove from the root canal.[112,115] In addition, it has been reported that zirconia posts are manufactured with a wide range of defects, resulting in some posts with very low fracture resistance.[118] Adhesive cementation of a ceramic post within the root canal is problematic because of the difficulty of bonding to the post surface.[119]

Hydrofluoric acid will not etch the zirconia surface, and air abrasion has variable effects depending on the resin cement used.[110] A variety of chemical primers have been developed and have been shown to enhance the bond to zirconia.[120–122] However, it appears that the bond created with these agents may be susceptible to degradation with aging.[123] Bond strength has also been shown to be improved by tribochemical coating$_{124}$ and by tribochemical coating plus a proprietary monomer.[125] Because of the susceptibility of zirconia posts to fracture, the increased risk of root fracture, and the difficulties associated with bonding, they cannot be recommended for routine clinical use.

Fiber-reinforced composite posts

Carbon FRC posts

Restoration of endodontically treated teeth with adhesively cemented FRC posts and resin-based composite restorative materials has become increasingly popular in recent years. The trend began in the early 1990s with the introduction of the carbon fiber post. One of the early carbon fiber posts (C-Post, Bisco) consisted of carbon fibers arranged longitudinally in an epoxy matrix. The carbon fiber content constituted approximately 64% of the bulk of the post.[126] It was marketed on the basis of its high strength, its compatibility with adhesive bonding techniques, and a modulus of elasticity comparable with that of dentin. It was postulated that this similarity in stiffness would reduce stress concentration and therefore prevent root fracture.[127] It appeared that the design was effective in this regard. In a 10-year retrospective study, the C-Post showed a success rate of 93%, with no failures due to root fracture.[128] A perceived disadvantage of carbon fiber posts was poor esthetics because of their dark color. With the advent of the glass and quartz FRC posts, carbon FRC posts lack a specific indication and are seldom used.

Glass and quartz FRC posts

Composition. Current FRC posts are fabricated of glass or quartz fibers embedded in a matrix of epoxy or resin composite. When this newer generation of FRC posts was introduced, they were marketed on the basis of improved esthetics because of their white color and translucency. The carbon fiber posts of the previous generation occasionally caused an objectionable dark shadow under highly translucent ceramic restorations. Today, most routinely used ceramic systems utilize a core of opaque

alumina, zirconia, or lithium disilicate, so the purported esthetic advantage of fiber posts has become less important. However, the potential advantages of a favorable modulus of elasticity and bonding compatibility remain.

FRC posts can be made of different types of glasses. Electrical glass (E-glass) is an amorphous structure composed of silicon dioxide (SiO_2) and other oxides and is the most commonly used type, although high-strength glass (S-glass) is also used. Additionally, some posts are made of quartz—pure silica in a crystalline form.[129] The fibers are often silanated to improve adhesion to the matrix material. They are manufactured by placing a fiber bundle under a load (*pretensing*) and then impregnating it with a liquid resin, which is then cured, often with heat. The load is then released, leaving the fibers under tension, which places the surface of the cured resin in compression to improve resistance to flexural forces.[130] The material is then cut and machined to the desired shape to produce the FRC post. A scanning electron micrography (SEM) image of the head of an FRC post illustrates the surface left by the machining process (Fig 21-14).

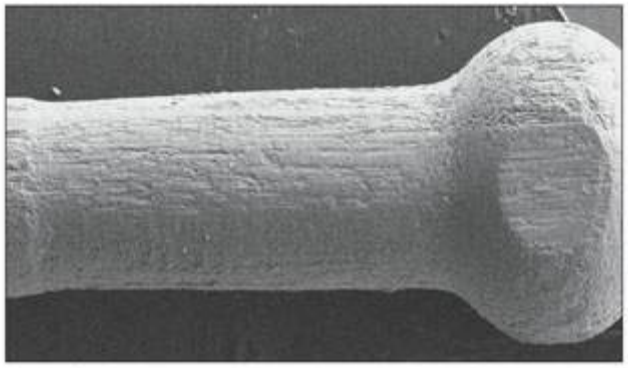


Fig 21-14 Surface characteristics of a machined FRC post are shown in an SEM image (original magnification ×20).

A typical FRC post (DT Light-Post, Bisco) consists of pre-tensed unidirectional 8-μm silanated quartz fibers in an epoxy matrix. The fibers occupy 60% of the volume of the post at a density of 32/mm^2.[75] In contrast, the matrix of another commonly used post (FRC Postec Plus, Ivoclar Vivadent) consists of triethylene glycol dimethacrylate (TEGDMA) and urethane dimethacrylate (UDMA) resins with dispersed silica fibers.[131] A wide variety of FRC post designs are available (see Fig 21-9b).

Strength and fracture resistance. It is difficult to directly compare the strengths of individual FRC post designs because of differences in surface topography (machined retentive features) and tapering, which prevent use of common flexural strength–

testing methods. However, Stewardson et al[132] tested the flexural properties of post source materials of consistent diameter and length according to International Organization for Standardization (ISO) specification (3597-2:1993, Method 1008B:1996) and determined that the strength of some commonly used fiber post materials exceeded that of gold and stainless steel. FRC posts from different manufacturers vary widely in their ability to withstand fatigue loading, most likely because of differences in manufacturing technique, shape of the post, and fiber density.[130]

Exposure to water and thermocycling has a variable effect on the strength of FRC posts over time, but it is doubtful whether the changes noted are of sufficient magnitude to cause clinical concern.[129,133] Additionally, when sealed inside the root canal, water storage for a 12-month period did not decrease the strength of fiber posts over that of the dry control.[134]

The literature is replete with studies conducted on the fracture resistance of various FRC post and core systems, with many conflicting opinions. Fokkinga et al[20] conducted a structured analysis of the literature on in vitro failure loads and failure modes of fiber, metal, and ceramic post and cores. The structured analysis process for in vitro data is similar to the systematic review, a widely accepted method used to evaluate and summarize the results of randomized controlled clinical trials. An initial MEDLINE search located 1,984 possibly relevant studies. After a series of more rigorous selection criteria were applied, 12 papers were eventually selected, and a multinomial statistical analysis was applied. Most studies described failure loads and modes resulting from static load testing of single-rooted teeth. In general, the prefabricated metal posts were found to have the highest failure loads and the ceramic the lowest. However, FRC posts exhibited more clinically favorable failures with a lower incidence of root fracture. In one study, loading a tooth restored with a steel post resulted in a 30% incidence of root fracture compared with none for the FRC post groups.[78]

Studies performed with restored extracted teeth are difficult to conduct and interpret because of the great number of variables that must be controlled. High standard deviations are often seen in these studies.

Protection of the root. There are many modes of failure of endodontically treated teeth—caries, periodontal disease, failure of endodontic treatment, or debonding of the post or crown, among others. However, the most catastrophic failure is root fracture. Root fracture often renders a tooth nonrestorable and results in extraction. Figure 21-15 illustrates the effect of the presence of a post schematically. A rigid

metallic post transmits a laterally applied occlusal force primarily to the cervical area and to the apex of the post. FRC posts have been proposed as a means of reducing the incidence of root fracture by distributing the forces throughout the root. This effect is theoretically achieved through a better match of modulus of elasticity between the dentin and the post.

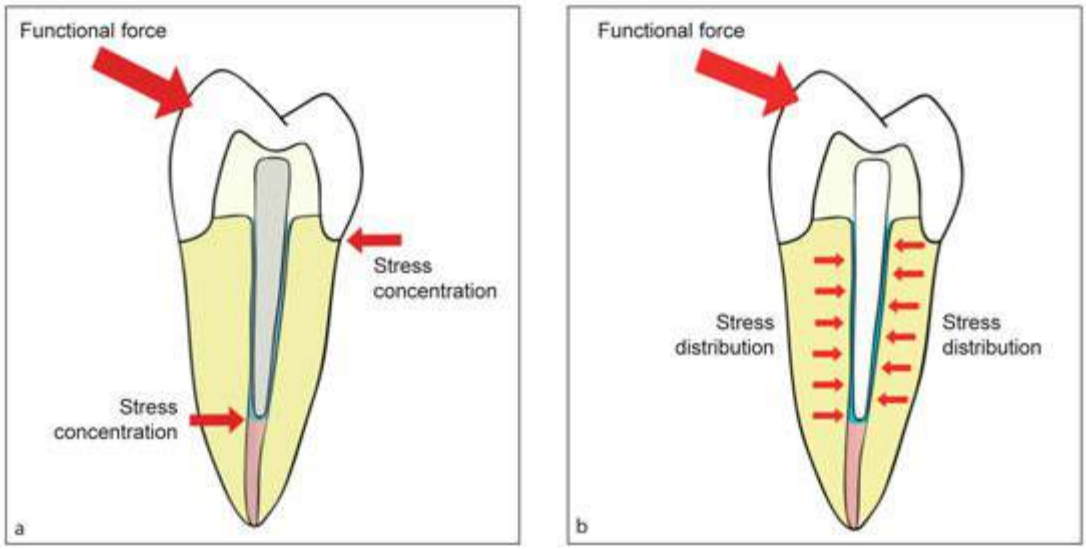


Fig 21-15 Comparative stress distribution within the root: metal post *(a)* versus FRC post *(b)*.

The observed modulus of elasticity of dentin varies depending on testing methodology and on the characteristics of the substrate tested. Regional variations of this dentin modulus also occur, depending on the number and diameter of tubules present. The bulk modulus of dentin has been reported to be in the range of 20 to 25 GPa.[135] Of the materials available for the construction of posts, quartz or silica FRC posts provide the best combination of strength and favorable modulus (Table 21-1).

Table 21-1	**Flexural strength and modulus of restorative materials and dentin**		
Material	**Type**	**Elastic modulus (GPa)**	**Flexural strength (MPa)**
Stainless steel[132]		194	743
Carbon fiber[132]	Composipost*	117	1,394
Titanium[132]		110	1,478
Gold[132]	Type IV	87	355
E-glass fiber composite[132]	Postec†	44	1,215
Quartz glass fiber	Light-Post‡	42	1,131

composite[132]			
Dentin[135]		16	200
Resin composite[136,137]	Z100§	20	139
Resin cement[138]	RelyX Veneer§	8	118
Ceramic[139]	Feldspathic porcelain	70	100

* RTD.
† Ivoclar Vivadent.
‡ Bisco.
§ 3M ESPE.

Predicting clinical performance with in vitro testing is difficult because of the complexity of the system being studied. Success depends on many factors: amount of remaining tooth structure, characteristics of the dentin, quality and effect of endodontic treatment, quality of adhesive bond, and others. Most laboratory studies test only one aspect of restoration behavior.

Finite element analysis (FEA) is a mathematic modeling technique that can be used to calculate stress distribution in structures made from materials with differing moduli and strengths. When applied to endodontically treated teeth, the results have been equivocal. A number of studies have found that use of posts with a stiffness similar to that of dentin results in better stress distribution within the root,[140–142] while others have reported higher stresses in roots restored with lower-modulus posts.[143,144] FEA techniques assume that the materials are homogenous and linearly elastic and that all interfaces are perfectly bonded—conditions that likely do not exist intraorally. Accurate depiction of true clinical conditions would require a level of sophistication not yet seen in the dental literature.

In vitro data tend to support the view that FRC posts have a protective effect on the root compared with metallic posts, but clinical evidence is limited. A Cochran Review published in 2007 identified only one clinical study that achieved the objective of comparing metal with nonmetal posts and also met the data quality criteria.[145] In this retrospective study, Ferrari et al[126] compared the outcomes of restoration with cast gold post and cores with that of FRC posts after 4 years of clinical service. The FRC post group exhibited a success rate of 95% with no root fractures. Clinical success in the cast gold group was 84% with a 9% incidence of root fracture.

Effect of ferrule. As discussed earlier in this chapter, the amount of ferrule, or

vertical height of the residual dentin above the finish line, is critical to clinical success of endodontically treated teeth. In recent years, a body of literature specific to the effect of ferrule on the performance of teeth adhesively restored with FRC posts has emerged. As with teeth restored with metallic posts conventionally cemented, it is clearly evident from these studies that success of endodontically treated teeth adhesively restored with FRC posts is strongly dependent on the amount of remaining tooth structure.[146] Regardless of restorative methodology used, this factor remains the most consistent predictor of success.[147]

Fatigue testing of extracted teeth restored with FRC posts and ceramic crowns emphasizes the importance of ferrule height. Teeth without a ferrule withstood only 212 load cycles; the presence of a 0.5-mm ferrule raised the mean failure to 155,137 load cycles; and teeth with a 1.0-mm ferrule survived the testing to completion at 250,000 cycles.[148] While it is ideal to have 360 degrees of circumferential axial wall dentin, when there are only partial walls remaining, the location of those walls may affect the prognosis of the restored tooth. As might be predicted by the functional forces applied, the presence of a palatal wall of dentin was found to be critical to the ability of anterior teeth restored with FRC posts to resist crown dislodgment.[149]

The importance of the amount of remaining coronal tooth structure is revealed in the results of clinical studies. In a clinical study of teeth that had been restored with oval-shaped glass fiber posts, Signore et al[150] found that failure increased dramatically when only one or two residual dentin walls greater than 1 mm in height were present. All debonding and core fracture failures were in these groups. In a 3-year clinical study, teeth with a ferrule less than 2 mm in height exhibited a failure rate of 26%. Teeth with a ferrule greater than 2 mm in height had a failure rate of only 7%.[151]

Perhaps the most convincing demonstration of the effect of remaining tooth structure is found in the clinical study by Ferrari et al.[24] A total of 360 teeth were classified according to the number of remaining walls prior to treatment with an FRC post, a resin composite core, and porcelain-fused-to-metal crowns. Examination of the results (Table 21-2) reveals a steadily increasing number of failures as the number of remaining walls decreases. This study also demonstrates the contribution of post placement to clinical success. The post does, in part, compensate for a lack of remaining tooth structure, in contrast to previously accepted thought.[152] The Ferrari study also suggests that the presence of a bonded FRC post may provide some degree of reinforcement of the root, as evidenced by the

observation that teeth without a post exhibited a higher incidence of root fracture than those with a post.[24]

Table 21-2	Clinical success rates of teeth without a post and with an FRC post according to the amount of remaining tooth structure after 6 years in service[24]					
	4 coronal walls	**3 coronal walls**	**2 coronal walls**	**1 coronal wall**	**No walls, ferrule present**	**No walls, ferrule absent**
No post	**100%**	**67%**	**53%**	**29%**	**11%**	**0%**
Crown debond		3	5	6	9	9
Endo failed		3	2	3	3	4
Root fractured			1	3	4	7
DT Light-Post	**100%**	**94%**	**89%**	**78%**	**61%**	**39%**
Post debond		1		3	4	4
Endo failure				1	2	3
Post fracture			1		1	3
Root fracture			1			3

Endo—endodontic treatment.

The dimensions of the ferrule and the number of remaining walls may be considered as primary predictors of clinical success. The previously cited clinical studies make it clear that when a circumferential ferrule of at least 2 mm is not present, the chance of failure increases dramatically.

Adhesion Within the Root Canal

A stated goal of adhesive restoration of endodontically treated teeth is to create a "monoblock" of perfectly bonded components of the same or similar modulus of elasticity—a "structurally and mechanically homogenous complex with dentin."[153] However, as we can see from the summary of clinical studies (Table 21-3), the bond

of resin cement to dentin in the root canal is far from perfect. The predominant failure mode of teeth restored with bonded FRC posts is consistently debonding of the post/restoration complex.

Advances in adhesive bonding to coronal dentin have led to materials and techniques that are durable and predictable, particularly when three-step etch-and-rinse adhesives are used.[158,159] When measured using comparable methodologies, the bond to root canal dentin appears to be lower than that to coronal dentin.[160,161] There are many reasons why this might be so: There are differences in the characteristics of root versus coronal dentin; there are possible effects from endodontic treatment; the root canal presents a geometry that results in the development of high stresses during resin polymerization; and access and visibility are poor for cleaning and bonding procedures.

Table 21-3 **Clinical studies for glass or quartz FRC posts cemented using an etch-and-rinse adhesive technique***

Authors	Mean study period (Y)	Coronal structure remaining	Teeth treated/ recalled	Tooth type	Definitive restoration	CF (%)	PD (%)	EF (%)	RF (%)	PF (%)
Malferrari et al 2003[154]	2.5	NR	180/NR	All	All-ceramic or metal-ceramic full crown	2	1			
Monticelli et al 2003[155]	2	Two coronal walls	225/NR	Pre-molars	All-ceramic full crown	6	4	3		
Grandini et al 2005[130]	2.5	50% residual tooth structure	100/NR	All	Direct composite	1		1		
Cagidiaco et al 2007[156]	2	0–4 walls	162/162	All	Crowns, direct composite	7	4	3		
Schmitter et al 2011[101]	5	At least 40% of crown destroyed	50/39	NR	Single crowns, FPD or RPD retainers	28	20	3		
		1–4 walls,			Metal-					

Ferrari et al 2012[24]	6	ferrule, no ferrule	120/101	Pre-molars	ceramic full crown	23	11	7	1	
Naumann et al 2012[157]	10	None vs one or more walls	157/1499	All	All-ceramic/ metal-ceramic full crown	37	11	5	3	11

NR–not reported; FPD–fixed partial denture; RPD—removable partial denture; CF–clinical failure; PD–post de
EF–endodontic failure; RF–root fracture; PF–post fracture.
*Numbers may not total correctly due to rounding error.

As a bonding substrate, root dentin is similar to coronal dentin. It was once thought that endodontic treatment led to a decrease in moisture content of the dentin.[3] This assumption has been challenged in recent years, and current thought is that endodontically treated teeth are very similar to untreated teeth in moisture content,[162] mechanical properties, and collagen cross-linking.[163] Some differences do exist. Structurally, root dentin has fewer tubules than coronal dentin and may exhibit tubular sclerosis, particularly in the apical region.[164–166] While these dentin factors may indeed affect in vitro bond strength,[167] with the current state of the art they are probably overshadowed by clinical technique and material issues.

Adhesion within the root canal is typically measured by three different techniques: pull-out tests, microtensile tests, and push-out tests. *Pull-out testing* involves cementing a post within the root of an extracted tooth and then fixing the root into position, gripping the post in the vise of a tensile testing machine, and pulling it out. The force required to remove the post provides an average measurement of the post retention. The pull-out test is difficult to use for FRC posts because the vise grip often crushes the post. *Microtensile testing* is often used for bond strength testing of coronal dentin and can be applied to root dentin. One difficulty with this technique is that the bonded interface is curved because of the shape of the canal, which may make interpretation of the data problematic. The *push-out test* is commonly used to evaluate bond strength within the root canal. With this technique, the FRC post is bonded within the canal of an extracted tooth, and the root is then sectioned into slices perpendicular to its long axis. The cross section of the cemented post within each slice is then loaded to failure, and the bond strength is calculated from the measured surface area of the debonded segment. The push-out test allows determination of bond strength at different levels within the canal and has been found to be more predictable than microtensile testing in this application.[161] See chapter 9

for more on adhesion testing.

Using the push-out methodology, a number of researchers have reported regional variability in bond strength in the root canal. In general, the bond strength decreases from coronal to apical sites. A consistent decrease in the length and density of resin tags formed during bonding procedures from the coronal to the apical aspect of the canal has also been noted.[168–170] This may be because of the aforementioned increase in tubular sclerosis in the apical region or regional differences in the effectiveness of bonding procedures because of decreased access and visibility.

Effects of polymerization shrinkage and stress

Resin composites shrink approximately 2% to 6% during polymerization.[171] This shrinkage results in contraction forces that may exceed the bond strengths of dentin adhesives.[172] This developing stress may be relieved when there are sufficient free surfaces compared with bonded surfaces in the restoration. The ratio of the bonded to nonbonded surface area is termed the *configuration factor*, or *C-factor,* and is commonly used to describe the severity of the stress challenge on the dentin bond.[172,173] For example, a Class 3 resin composite restoration would have a C-factor of about 1, while a Class 1 restoration might have a C-factor of close to 6. In the root canal, the cement is bonded to an extremely large surface area of dentin and FRC post in comparison with the amount of free surface. It has been estimated that the C-factor of the root canal system may exceed 200.[160] This high-stress state may result in areas of debonding and formation of gaps in the bonded interface. These gaps decrease the overall bond strength occurring within the canal relative to that possible when bonding to a flat surface.[160] The formation of gaps is exacerbated by the conditions of moisture and repetitive occlusal loading found in the oral cavity. The effect of moisture was investigated by Bonfante et al,[174] who found that the percent of continuous dentin interface decreased with 3 months of water storage. Gaps were initially observed in the apical and middle thirds, but these had extended to the cervical area by 3 months. Teeth are continually subjected to fatigue loading in function, and this too can affect the integrity of the bonded dentin interface. Push-out strength decreased for all adhesives tested following thermomechanical loading.[175] The formation of gaps following fatigue loading is dependent on the post material, with FRC posts showing fewer gaps than metallic posts, presumably through the mechanism of better stress distribution.[176] Mannocci et al[177] examined the resin-dentin interface of teeth restored with FRC posts that had been extracted for periodontal reasons after periods of 6 months to 6 years of function. Debonding was

observed in approximately one-third of the evaluated interfaces. Cyclic fatigue forces are likely a major factor in the observed high proportion of debonding among clinical failures of restored endodontically treated teeth.[178]

Effects of post length and diameter

It is clear that geometric, material, and technical issues result in an imperfectly bonded interface between cement and dentin in the root canal. The relative contribution to post retention of mechanical bonding through hybridization versus purely frictional effects is a subject of debate in the literature.[179–181] It has been previously noted that lower bond strength is found in the apical region than in with the coronal region. Nonetheless, increasing post length does seem to increase post retention.[182,183] Therefore, the principle of increasing length to increase frictional retention, first recommended for metallic posts,[85,92] is also applicable to FRC posts.[141] As a general guideline, Sorensen and Martinoff[184] recommended that the post length should at least equal the crown length. The cervical region of teeth restored with posts is subjected to the highest stress levels,[185] and greater post length allows better distribution of these stresses along the root. When an anterior tooth is experiencing functional forces, the point of approximation of the cervical root with the crest of bone acts as a fulcrum point. Therefore, a more conservative guideline is that the length of the post should be greater than the length of tooth structure extending above the osseous crest.

Three general guidelines may be stated for determining post length:

1. Tooth structure should be preserved. Excessive radicular dentin should not be sacrificed to extend the length of the post.
2. At least 4 mm of gutta-percha should be left apically.
3. The length of the post should be at least equal to the amount of tooth coronal to bone.

Increasing mechanical retention by roughening canal walls with a diamond bur[186] or using specially designed burs to create undercuts in the canal has been proposed, but these techniques are not recommended because removal of additional tooth structure may increase the fracture potential of the root.

It has been reported that increasing the diameter of metallic posts does not significantly contribute to the frictional resistance of the post to displacement[85,187] and significantly increases internal stresses within the tooth.[188,189] In relation to

bonded FRC posts, increasing post diameter may have additional deleterious effects. As the diameter increases, so does the load-bearing capacity of the post. For example, a ¼-mm-diameter FRC post was found to have a load-bearing capacity of 85 N, whereas a 2.1-mm-diameter post of the same type has a capacity of 200 N. This increased load-bearing capacity due to increased diameter results in increased rigidity of the post system, which could override the elastic modulus, which is a material property.[20,129] Thus, an increased diameter of post relative to the remaining dentin wall thickness could result in root fracture as a common failure mode, similar to that observed in stiffer metallic or ceramic posts.

Clinical Principles of Adhesive Restoration

Clinical procedures should emphasize preservation of tooth structure, protection of the root, support of the crown, and augmentation of retention and resistance form with adhesive bonding. The first principle, conservation of tooth structure, begins with the endodontic procedure. A minimum of tooth structure should be sacrificed for access and instrumentation of the canal. A post should be placed only when required, because even with careful clinical technique, post space preparation results in the removal of tooth structure, increases the risk of root perforation, and increases time and expense. Optimally, the post should fit the canal as it presents following endodontic treatment. The canal should be occupied to the greatest extent possible with FRC post material, because it is stiffer and stronger than resin-based cements. Similarly, the core should be comprised as much as possible of FRC post material, because its higher stiffness compared with resin composite provides better support for the overlying crown material. An increased core modulus results in greater fracture resistance of ceramic crowns.[138,190,191] The measured modulus of elasticity of an FRC post increases as the load application progresses from 90 degrees to the long axis (as measured and reported in Table 21-1) to a more compressive load as the load becomes normal to the fibers of the post. Therefore, because of the geometry of the post and the direction of occlusal forces, the post material is much stiffer and stronger than the resin composite core material in which it is embedded. Consequently, the FRC post should be left at full length through the placement and buildup procedures and then cut to appropriate length during tooth preparation for the crown. In this way, the FRC post extends to the occlusal or incisal cement layer and provides maximum stiffness and support.

Table 21-1	Flexural strength and modulus of restorative materials and dentin

Material	Type	Elastic modulus (GPa)	Flexural strength (MPa)
Stainless steel[132]		194	743
Carbon fiber[132]	Composipost*	117	1,394
Titanium[132]		110	1,478
Gold[132]	Type IV	87	355
E-glass fiber composite[132]	Postec†	44	1,215
Quartz glass fiber composite[132]	Light-Post‡	42	1,131
Dentin[135]		16	200
Resin composite[136,137]	Z100§	20	139
Resin cement[138]	RelyX Veneer§	8	118
Ceramic[139]	Feldspathic porcelain	70	100

Therefore, the ideal post design includes a cylindric coronal portion and a conical apical portion.[192] FRC posts that are ovoid in cross section have been developed to better fill the space left by endodontic instrumentation of noncircular canals.[193] Used with ovoid ultrasonic post space preparation instruments, these posts have been found to provide clinically acceptable performance.[150] Conventional FRC posts may be ground to better fit the shape of the canal without loss of surface integrity.[194] However, the procedure is tedious and may be impractical in many situations. When presented with an overly flared canal, the clinician should strive to fill the space around the post with higher-strength/modulus resin composite versus resin cement. FRC "accessory posts" have been developed that may also prove to be useful.[195]

A minimal thickness of the cement layer should be achieved with a properly selected post shape and size. With decreased cement thickness, volumetric shrinkage is reduced.[196] The cement selected should have a low filler loading, because this will result in enhanced flow and adaptation and less shrinkage stress development in the confined canal space compared with more highly filled materials.[197,198]

To summarize, different material requirements exist in different areas of the tooth/restoration construction. The ideal structure would include an FRC post that fills the canal and the space occupied by the core to the greatest extent possible, a

minimal cement thickness, and a high-strength, high-modulus core material (Fig 21-16).

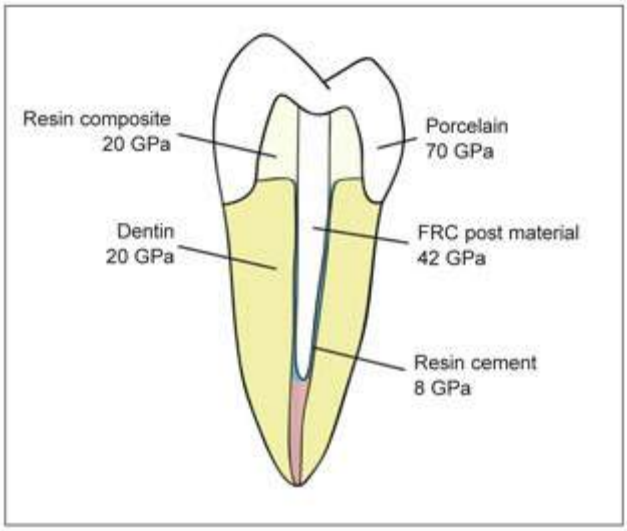


Fig 21-16 A comparison of moduli of elasticity of current restorative materials with that of dentin. The ideal structure would include an FRC post that fills the canal and the space occupied by the core to the greatest extent possible, a minimal cement thickness, and a high-strength, high-modulus core material.

Canal preparation

The root canal dentin of teeth that have been endodontically treated may have been exposed to materials and chemicals that have the potential of altering the dentin structure or interfering with resin polymerization. The canal may have been rinsed with an irrigant such as sodium hypochlorite, hydrogen peroxide, or ethylenediaminetetraacetic acid (EDTA). The dentin may have had extended exposure to sealers or calcium hydroxide. This history must be factored into the canal preparation and selection of adhesive materials.

The inhibitory effect of eugenol on resin polymerization is well known.[139] Eugenol readily diffuses from materials that contain it into adjacent dentin.[199] Eugenol-containing sealers are commonly used in endodontic treatment and have been implicated in decreased bond strength of resin-based adhesives,[200,201] through some researchers have found no effect.[202,203] The key factor seems to be mechanical preparation of the root canal prior to post cementation. Research indicates that this eliminates the effect, presumably by removing the eugenol-contaminated dentin.[204] The use of adhesive techniques incorporating phosphoric acid etchant also appears to restore bond strength.[205,206] The timing of canal preparation also is of concern. Endodontic sealers may exhibit extended setting times. McComb and Smith[207]

reported the setting time of Roth's 801 (a eugenol-containing material) to be 3.5 hours; the setting time of AH26 (a resin-based, noneugenol sealer) was 10.5 hours. In the case of post space preparation and cementation immediately following endodontic obturation, it may be difficult to clean the unset sealer from the canal walls. Also, even when walls appear to be clean, sealer may be inadvertently spread from the area of the apical seal during the application of adhesive agents via microbrush or paper points. Vano et al[208] reported decreased bond strength with a variety of adhesive techniques when canals were prepared and FRC posts cemented immediately following obturation. When post preparation and cementation was delayed by at least 24 hours, the bond strengths were statistically equivalent with that observed in unobturated roots. Clinicians should exercise caution when using a resin luting agent to cement a post immediately following canal obturation with eugenol-containing sealers.

A canal preparation drill is typically included with FRC post kits, and often the manufacturer's instructions specify preparation limited only to that instrument. Because the drill cross section is circular, and because canals are seldom round, use of this instrument exclusively will undoubtedly result in gutta-percha and sealer remaining in the recesses of oval- and ribbon-shaped canals unless the canal is overprepared. Therefore, canal preparation to receive a bonded FRC post should begin with smaller instruments such as Gates-Glidden burs. The flexibility of the Gates-Glidden bur allows the removal of gutta-percha with minimal risk of perforation. Creation of clean canal walls suitable for bonding can be surprisingly difficult. Even with the use of Peeso reamers prior to use of the post drill, a significant amount of debris remains on canal walls, particularly in the apical region.[209] The presence of this debris results in incomplete hybridization of the dentin in the contaminated areas.[210]

A post should be selected to fit the space created during endodontic treatment without significant further alteration of the canal. The sizing drill should only be used to remove any minor amounts of dentin that act as an impediment to the complete seating of an appropriately selected diameter of post. It should not change the existing anatomical structure of the canal as it presents postendodontically. Heavy apical or lateral forces should be avoided with the post drill. If the canal is properly prepared with the previous instruments, only a light force should be required. Meticulous technique is necessary in post space preparation, and a radiograph prior to cementation (without the post in place) is recommended to help detect any remaining gutta-percha or sealer.

Selection of bonding agent

A variety of materials and methods for bonding FRC posts are available. These include the traditional etch-and-rinse adhesives used in conjunction with resin cement, self-etching cements, and self-adhesive cements. For coronal dentin bonding, the three-step etch-and-rinse adhesive systems (the so-called fourth-generation bonding agents) involving the use of phosphoric acid etchant, hydrophilic primer, and adhesive remain the gold standard compared with simplified techniques.[158,159] The overwhelming majority of FRC posts in clinical studies were cemented using an etch-and-rinse technique; thus, it is the most clinically proven method. Although controversy exists, in vitro bond-strength studies comparing etching and rinsing with other cementation techniques in root canal dentin generally reflect the clinical results. The three-step etch-and-rinse adhesives are preferred for bonding in the root canal.[211–213]

Unfortunately, the use of etch-and-rinse adhesives within the root canal adds complexity and technique sensitivity to the post cementation process. Overcoming access and visibility constraints requires special techniques to achieve complete coverage of the walls with etchant and to ensure its removal. Similarly, primer application and solvent evaporation can be problematic. Effective etchant application requires the use of a cannula attached to the etchant syringe that will reach the apex of the canal preparation. The tips provided in the various FRC post kits are often inadequate for this purpose. The etchant is injected with a cannula of sufficient length while the tip is withdrawn. If a tip is used that does not reach the complete depth of the canal, a bubble will be formed apical to the point that the gel contacts the canal walls circumferentially, and the gel will go no further (Fig 21-17). If the canal is known to be deeper than the length of the tip, the gel may be carried to the apex with a microbrush. If gel is allowed to remain on the walls, primer penetration will be impeded, and a low-pH environment will be created that may be detrimental to the curing of the dual-curing adhesive that follows.[214] Potesta et al[170] demonstrated the superiority of this technique in a push-out study. The use of an appropriate cannula and technique resulted in higher bond strengths. An air-water syringe was ineffective even when used for periods up to 2 minutes. Rinsing the etchant with the same cannula and a syringe raised the pH level to neutral with as little as 5 ml of water.

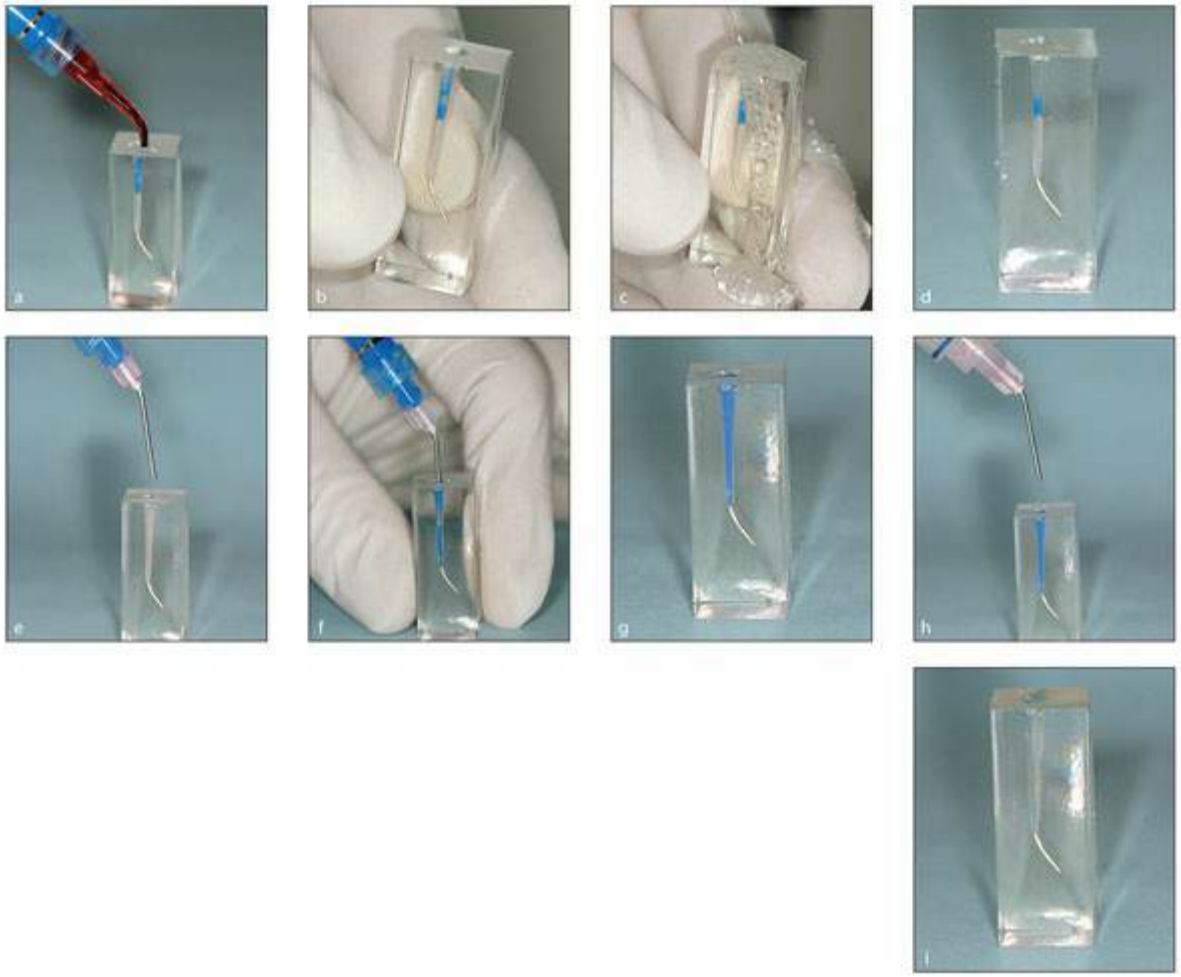


Fig 21-17*(a)* Use of a tip of inadequate length. *(b)* A bubble forms at the point to which the tip extends, and the etchant will go no further. A microbrush must be used to obtain complete coverage. *(c)* Rinsing with an air-water syringe results in incomplete removal of the etchant. *(d)* Etchant remains on the canal walls and will impede bonding. *(e)* A cannula of adequate length is attached to the etchant syringe (Endo-Eze, Ultradent). *(f)* The tip is inserted to the depth of the canal, and the etchant is injected while the tip is withdrawn. *(g)* Complete coverage of the canal with etchant. *(h)* The same tip is attached to an irrigating syringe. *(i)* Complete removal of the etchant.

Care must be taken to ensure uniform application of primer to the canal walls. This is best accomplished through the use of an appropriately sized microbrush.[215,216] There are a number of products on the market that are effective (Fig 21-18). Removal of excess and evaporation of solvent from the primer may be accomplished by repeated insertion and blotting of a microbrush or through the use of paper points. Application of air from the air-water syringe may aid this process. Great care must be used in this step. If excess solvent is allowed to remain, it may inhibit polymerization in the hybrid layer.[217]



Fig 21-18*(a and b)* Microbrushes for primer application and removal of excess.

A dual-curing adhesive is then applied using a microbrush, and the excess is removed. The dual-curing cement should be applied to the canal using a similar technique. Currently, some manufacturers include tips with their cement that attach to the end of the mixing tip and will reach the apex of most canals (Fig 21-19). If the length of the tip is determined to be insufficient, the cement may be injected into the back of a disposable syringe with a longer cannula attached for delivery into the canal (Fig 21-20).



Fig 21-19 Dual-curing cement with automix tip and 17-mm applicator tip (RelyX Ultimate, 3M ESPE).



Fig 21-20 Disposable syringe (Skini Syringe, Ultradent) with 22-mm Endo-Eze applicator tip.

Self-etching and self-adhesive cements are marketed as being more convenient to use for post cementation. With self-etching cements, a self-etching primer is applied to the dentin, and the mixed cement is applied to the canal without rinsing. Self-adhesive cements are applied directly to the canal without etching or the use of additional primers or adhesives. While varied results may be found in the literature, they are generally less effective than etch-and-rinse systems.[131,218,219] Low bond-strength values and the lack of long-term clinical data limit the application of recently marketed self-etching and self-adhesive materials for the cementation of FRC posts.[213]

Effect of tag length

In coronal dentin, the majority of the bond strength is derived through hybridization of the intertubular dentin, with a minimal contribution from resin tags formed within the tubules.[220] Within the root canal, the situation appears to be somewhat different. In push-out testing, well-defined tag formation is usually found in areas of high bond strength, while regions with lower adhesion typically exhibit a poorly structured interface.[170,176] Studies show a decrease in tag length and definition from the coronal to apical regions regardless of the adhesive system used.[167,170,210,221] However, the three-step etch-and-rinse adhesives seem to be more effective than simplified systems in creating well-defined resin tags in the apical and middle third.[211] Figure 21-21 illustrates the difference in resin tag formation among a three-step etch-and-rinse adhesive, a self-etching cement, and a self-adhesive cement.[222] Abundant resin tags are noted with the three-step etch-and-rinse adhesive, and it is conceivable that these tags could contribute to frictional retention of the post. In contrast, self-etching and self-adhesive cements exhibit a superficial interaction with root canal dentin. They are not as effective in removing or

penetrating the thick smear layer produced during canal instrumentation and therefore exhibit poorly defined tag formation.

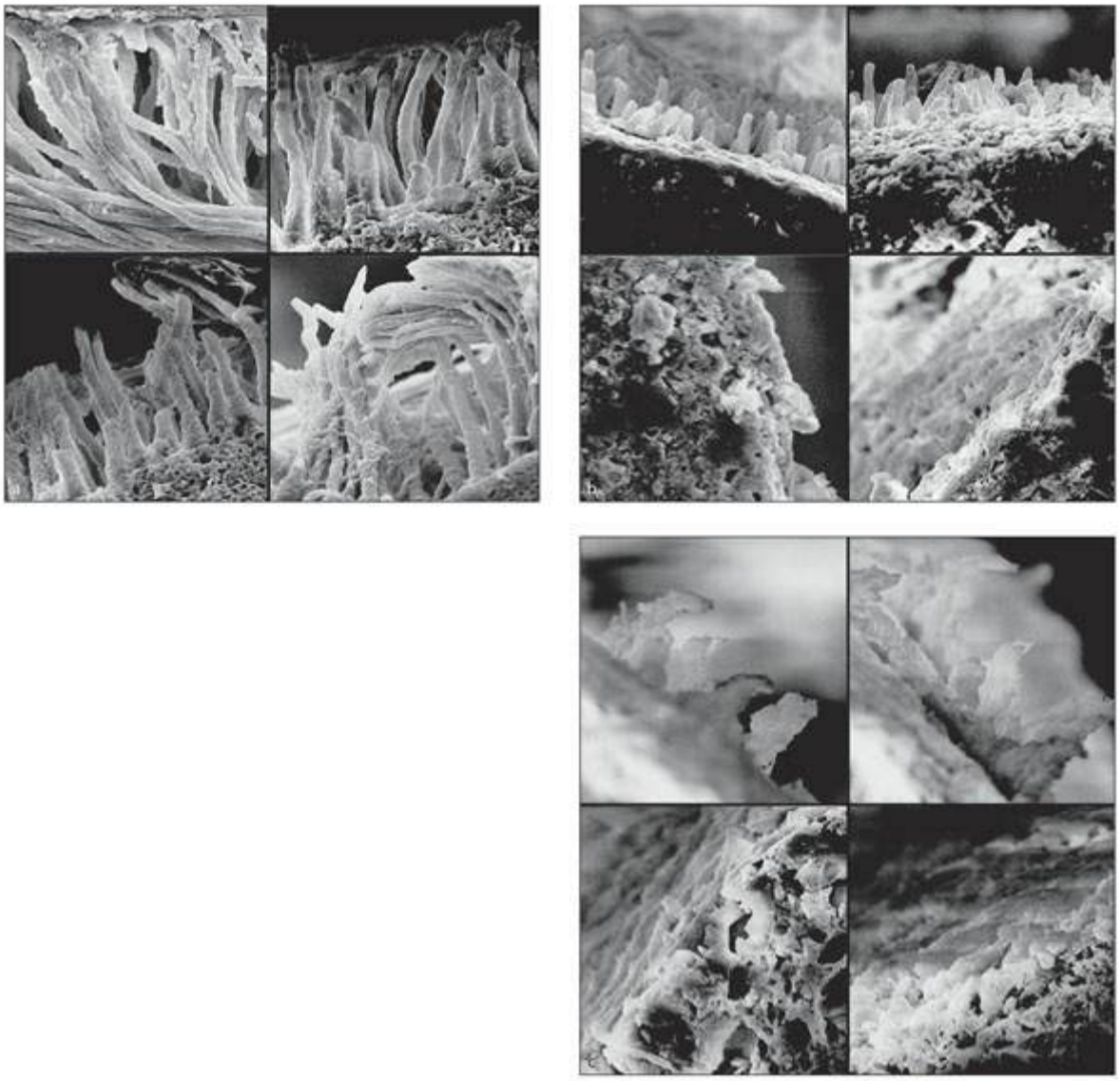


Fig 21-21 *(a)* Resin tag formation with a three-step etch-and-rinse adhesive (Permaflo DC Primers, Ultradent). *(b)* Resin tag formation with a self-etching cement (Panavia F 2.0 with ED Primer, Kuraray). *(c)* Resin tag formation with a self-adhesive cement (RelyX Unicem, 3M ESPE). *(clockwise from upper left)*: coronal dentin, two mid-root locations, apical region.

Bonding to the post

The matrix polymers used in FRC posts typically exhibit a high degree of conversion and are highly cross-linked.[129] The possibility of creating a chemical bond between the luting adhesive and the post is therefore low. However, machining of the post to the desired shape by the manufacturer results in a rough surface, which provides a mechanical interlock for the adhesive (see Fig 21-14).

A number of methods to improve adhesion to the FRC post have been recommended. Most are intended to remove matrix material from the post to expose the silica-containing fibers. A silane coupling agent is then applied in an attempt to achieve a chemical bond of the silica-containing substrate with the resin cement or resin composite. The use of air abrasion, etching with concentrated hydrogen peroxide,[208] sodium ethoxide,[223] and other agents have been suggested as a means of removing resin matrix from the post surface. Air abrasion must be used with caution, because it can quickly destroy the post if applied for too long. Used appropriately, it has no effect on the mechanical properties of the post.[224] Air abrasion has been found to improve resin retention, with or without silane.[225,226] Treatment of the FRC post with concentrated hydrogen peroxide for as little as 1 minute has also been shown to be effective,[227] as has application of silane to the unetched post surface.[228] The use of hydrofluoric acid$_{208}$ and tribochemical coating$_{229}$ have been evaluated as well but are somewhat cumbersome to use and may damage the surface of the FRC post. The literature regarding the various surface treatments is conflicting, and most are tested after only a short period of water storage and without being subjected to a thermomechanical challenge. Khamverdi et al$_{230}$ evaluated the effect of hydrogen peroxide etching compared with air abrasion on microtensile bond strength to a FRC post after water storage. A steady decrease in bond strength was observed over the storage period, and by 9 months there was no difference between the two treatments. Some post manufacturers recommend application of a bonding agent to the post surface prior to cementation. While not harmful, this has not been shown to increase bond strength.[231]

FRC posts with a machined macroretentive surface are available. While this feature undoubtedly increases mechanical retention at the post-cement interface, it results in a significantly lower fracture resistance than a similar sized post with a smooth surface.[78] Attempting to improve the adhesion beyond that provided by a clean post surface may not be clinically useful. Boschian Pest et al[168] determined the bond strength a variety of adhesives and cements to fiber posts and to dentin. With each combination, the bond to the post was significantly higher than the bond to dentin. The results of most push-out and pull-out bond studies indicate that failure is typically at the dentin interface rather than the post interface.

Box 21-2	**Armamentarium for FRC post cementation**
Restorative setup and routine disposables Boley gauge Post sizing drill	Etchant gel Cannula tip 5-ml syringe

Fiber post Resin composite restorative material Gates-Glidden burs (nos. 2 to 5) Silicone stops Alcohol wipe	Three-step etch-and-rinse dual-curing adhesive Dual-curing cement with mixing and injection tips Barrier sleeves Microbrushes (3)

Box 21-3	**Clinical steps for FRC post cementation**
1. Administer local anesthesia if necessary. 2. Estimate the length/diameter of the post using the final endodontic radiograph. 3. Isolate the tooth with a rubber dam. 4. Remove existing provisional and any remaining restoration. 5. Set the Gates-Glidden bur to the desired length with a silicone stop and remove the gutta-percha. 6. Set the post sizing drill (corresponding to the post selected) to the established length with a silicone stop and remove any minor obstructions to seating. 7. Take a radiograph to confirm complete gutta-percha removal. 8. Clean the post with an alcohol wipe; set the silicone stop to the established depth of insertion. 9. Attach a cannula tip to the etchant syringe, insert it to the full depth of the canal, and inject the etchant while withdrawing the tip. Coat all radicular and coronal tooth structure to be bonded with etchant. 10. Remove the cannula tip from the etchant syringe and attach it to a 5-ml-syringe loaded with water.	11. Irrigate the canal and surrounding tooth structure with water. 12. Dry with an air syringe; blot the canal with a micro brush. Leave the dentin slightly moist. 13. Apply primer to all dentin surfaces with a microbrush. 14. Evaporate solvent from the coronal dentin with a gentle stream of air. Remove excess primer from the canal by repeated blotting with microbrush. 15. Apply dual-curing adhesive to all dentin surfaces with a microbrush; use a bristle brush to ensure even coating and remove excess from the coronal surfaces. Remove excess adhesive from the canal by repeated blotting with a microbrush. Do not light cure. 16. Attach a new mixing/injection tip to the cement syringe; bleed a BB-sized amount of cement through the tip. 17. Insert the tip to the depth of the canal and inject while withdrawing. Do not overfill. Immediately insert the post and hold it in place with moderate pressure. The post should be inserted slowly to allow dissipation of hydrostatic forces. Remove any excess cement. 18. Light cure for 20 seconds. 19. Proceed with core buildup with resin composite.

Cement selection

Both light-curing and dual-curing resin cements are available for bonding of FRC posts. Techniques using light-cured adhesives and cements polymerized via light-transmitting posts have been suggested. FRC posts have been presumed to conduct light via the glass or quartz fibers, but these may in fact contribute a relatively small effect compared with the light-transmission capabilities of the resin matrix material.[232] Light-transmitting capabilities vary widely among different posts because of material variations. For all posts, light transmission decreases from coronal to apical. Some FRC posts have been found to transmit no light at all.[233] Relative post translucency affects the degree of conversion of the light-cured cement,[234] but even a highly translucent post exhibits a limited ability to transmit sufficient light to fully cure the resin. In a study assessing the light-transmission capabilities of a translucent fiber post, Galhano et al$_{235}$ found that the microhardness of the resin cement decreased with distance from the light source and that the apical third of the cement remained unpolymerized. Because of uncertainties regarding complete polymerization with the use of light-cured cements, dual-cured materials are recommended. The self-curing mode ensures polymerization at the depths of the canal, and the relatively slow set may allow some stress relief within the cement layer.[236,237] Light curing the dual-curing cement following initial set will stabilize the post to allow the clinician to proceed with the core buildup. It also increases conversion and improves the mechanical properties of the cement.[238–240]

Clinical Technique

From the preceding discussion, it is clear that clinically successful adhesive restoration of endodontically treated teeth requires a detailed understanding of the bonding substrate and materials and scrupulous attention to detail. A suggested armamentarium and procedure for FRC post cementation are shown in Boxes 21-2 and 21-3.

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